STEP-BY-STEP: ESTIMATE WHETHER A MODEL WILL FIT

Now examine the model.

- **Step 1: Find the exact file size or memory estimate** for the quantisation you want.
- **Step 2: Compare it with available VRAM** if you want full GPU inference. Leave headroom for the runtime and KV cache rather than filling every last megabyte.
- **Step 3: If it exceeds VRAM,** compare it with system RAM and check whether your runtime supports partial GPU offload.
- **Step 4: Start with a modest context window.** Do not select 128K merely because the model supports it.
- **Step 5: Load the model and observe actual memory use.** In Ollama, ollama ps shows whether execution is 100 percent GPU, 100 percent CPU or split between them.
- **Step 6: Run the same prompt several times and judge responsiveness.** A model that fits but frustrates you has failed the practical test.

side is that it shares bandwidth and power with the CPU, so performance may still lag behind a discrete GPU.

Treat integrated graphics as acceleration worth testing rather than guaranteed magic. If your runtime offers CPU-only and GPU-enabled modes, benchmark both.

How much hardware do you actually need?

Use the following as broad starting bands, not hard limits.

An **8GB RAM laptop** should focus on very small quantised models and short contexts. Close background applications, and expect CPU-oriented inference unless the system has a supported accelerator.

A **16GB mainstream laptop** is a reasonable starting machine. Small models are comfortable, and some mid-sized Q4 models become possible if you keep context sensible. This is also LM Studio's recommended RAM baseline.

A **32GB system** is where experimentation becomes much more enjoyable. You can keep larger models in system memory, try CPU-GPU splits and run ordinary applications alongside the model without immediately starving Windows.

A **gaming laptop with 8GB VRAM** can accelerate many small and medium quantised models very effectively. Do not assume that a powerful RTX badge removes the 8GB capacity limit.

With **12-16GB VRAM**, you enter much more flexible territory. More of a model can remain entirely on the GPU, and you have greater room for context.

At **24GB VRAM**, serious local inference becomes realistic. Mistral, for example, recommends a 24GB RTX 4090-class GPU for a four-bit Mistral Small 4 setup at around 32K context. That is a useful reminder that even 24GB is not infinite when total model sizes reach triple-digit billions of parameters.

Finally, **large-memory Apple Silicon machines** occupy their own category because unified memory can allow GPU access to capacities that would require expensive workstation GPUs on a conventional PC.

What should you upgrade first?

If your machine already runs the models you need, upgrade nothing.

If models fail because system memory is exhausted and your laptop or desktop supports RAM upgrades, moving from 16GB to 32GB can be one of the most cost-effective improvements.

If you want dramatically faster generation and your workloads fit inside a discrete GPU's memory, a stronger GPU can transform inference. When comparing GPUs for this purpose, put VRAM capacity much higher on your checklist than you would for ordinary gaming.



If you are buying a non-upgradable machine such as many thin laptops or Macs, buy enough memory at the start. Local AI has a remarkable ability to make "I'm sure 16GB will be enough" age badly.

Finally, resist the urge to build a monster machine merely to run the largest model you can find. The objective of local AI is not to win a specification contest. It is to make useful AI available quickly, privately and reliably on hardware you control.

The right question is therefore no longer, "Can my computer run AI?"

Almost certainly, yes.

The useful question is, "Which AI can it run well?"